

Project 3
EAE 135 Winter 2026

Deadline: Saturday March 14th, 2026 at 11:59 pm on Canvas
(if working in teams, upload only one submission)

Objectives

To familiarize yourself with the structural integrity of a light aircraft design; to analyze a fuselage cross section to determine its safety against shear stress caused by a representative concentrated force.

Scenario

There are various loads applied to an aircraft fuselage and it needs to be designed to withstand all these loads. There are loads that can result in axial stresses, such as loading resulting from the cabin pressurization, or axial stress due to bending. Some loads can result in shear stress, particularly shear stress acting on the plane of fuselage cross-section, which is of interest in this analysis.

Students will be analyzing a representative fuselage cross-section made of Aluminum 6061-T6. A relatively large off-center concentrated load is applied in the vertical direction. Assuming the fuselage to be sufficiently long and considering the skin has a much smaller thickness compared to cross-section length scales, we analyze the cross-section according to simplifications offered by the assumptions of a thin-walled 3D Euler-Bernoulli beam. Further simplifications for this analysis shall be applied: assume the cross-section is idealized by separating the functions of stringers and skin panels, assign axial loads only to the former and shearing loads only to the latter. The fuselage weight will be neglected and the elastic modulus is assumed to be the same in tension and compression.

In this analysis, you (and your team) shall calculate the contributions of shear flows due to bending in the skin elements.

From there, we can calculate shear stress in each element and thus assess the safety of the cross-section in shear.

Finally, discuss a few ways to strengthen the cross-section against shear if necessary.

Figure 1 represents the geometry of the idealized cross-section and the applied concentrated force V_3 . The curved portions at the top and the bottom are semi-circles with radius R and the straight lines ($\overline{23}$ and $\overline{56}$) have a length R .

Stringers 1, 2, and 6 have area A_1 and stringers 3, 4, and 5 have area A_2 .

The skin has a thickness t everywhere.

Force V_3 is applied at a distance b in the negative x_2 direction from point O . **Table 1** provides parametric values which correspond to the problem statement. If making a cut, do so along skin section 12 and move counter-clockwise from this location.

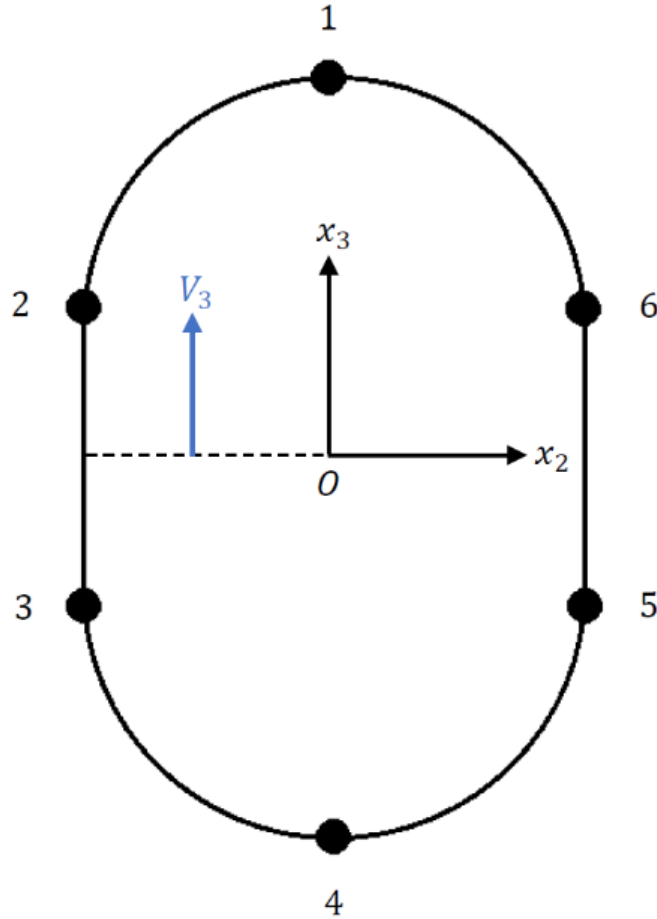


Figure 1. Fuselage cross-section and loading model

Table 1. Parameter values

Parameter	Value	Parameter	Value
R (mm)	500	V_3 (kN)	80
b (mm)	400	E (GPa)	68
A_1 (mm ²) for stringers 1, 2, 6	120	G (GPa)	26
A_2 (mm ²) for stringers 3, 4, 5	150	(A basis) τ_{su} (MPa) Ref: MIL-HDBK-17	186
t (mm)	2	SF	1.5

Miscellaneous Information

You can work in teams up to 3 people. The submitted report should be professional and easily readable. Typing is best, but neat handwriting is acceptable. We are asking for the results to be in tables.

The recorded presentation format is up to you (e.g. use of a board, PowerPoint, tablet, single presenter or group of presenters), but we do expect professionalism. You are not required to dress up but we expect you not to be in a swimming attire or in a PJ.

Once the project grades are posted, each team or individual will have a 24 hours window to contact the instructor of record asking for clarifications or making a case (based on the grading rubric) for a regrade.

AI Policy

Students are not allowed to use advanced automated tools (artificial intelligence or machine learning tools such as ChatGPT) on any assignment of this course.

Resources for this Project

- See Example 8.26 from Bauchau and Craig.
- File OLD_EXAMPLE_Project3_EAE135_W17.pdf
- Video recording on Canvas Media Gallery (lecture15_part2 and project 3 overview)

Deliverables

You are asked to submit **three items**.

Item 1: a professional report answering the following questions:

1. Calculate the total shear flow in the fuselage section due to the applied loading. Show the steps and comment on your results. Present the total shear flow for each segment in a table.
2. How do you use the results from a) for a failure analysis of the fuselage?

Item 2: a recorded presentation. Include the link of the video inside the report. Check before submission to make sure the video is viewable.

The video time length is constrained to **5 minutes**. The face of the presenter(s) need(s) to be visible during most of the video.

The presentation needs to address the three aspects below.

- 1) Problem statement: what is the problem to solve
- 2) Solution method: what methods were used, and what are their hypotheses
- 3) Discussion of the results: how results are used, and do they make sense

The presentation will be graded also for 1) clarity and professionalism, 2) the ability to convince the instructor that the student(s) understand(s) the topic and did the work, 3) recording quality.

Instructions for uploading the video of the recorded presentation

- Go to video.ucdavis.edu

- After entering your login credentials (top right of your screen), click on top right corner, '+Add new'.
- Drag the video you recorded to the 'Upload media' field.
- Click on **Publishing status** → **select 'unlisted'**
- Enter description: your name, recorded presentation number #
- **Save**

After the video has finished uploading and you see green-fielded 'your changes have been saved', go back to your name on the top right corner, and click on My Media. You will see the video being processed/compressed. Click on the video. You will see **Details Share Symbol of a magnifying glass**. Click on 'Share'. You will see a link. **Copy and paste the link into your submission.**

The video must be checked prior to submission. The link must be checked to make sure it is working properly and can be viewed. In the past, people forgot to follow instructions.

There will be a 5% penalty if the video is not viewable and we need to contact you to upload it properly. Please do not erase the video until the quarter is over, in case we decide to go back and review it. Please **DO NOT** add us as collaborator/co-viewer.

Item 3: an appendix with references used, workload distribution (in a team of up to 3), any code used (**in text form and readable**). If you used a spreadsheet, upload it as a separate file.

Grading Rubric

EAE 135 Project 3, Winter 2026

Grading Rubric (100 point scale)

Authors: _____ || _____

Technical Requirements of Report

_____/40

- Equations with units where applicable _____/5
- Correct calculations of open shear flow _____/20
- Correct calculation of closing shear flow and torsional shear flow _____/10
- Safety analysis and recommendations _____/5

Delivery of Report

_____/30

- Clear rhetorical structure (thoughts and concepts are presented in an organized, logical structure) _____/10
- Quality of text (neat, professional) _____/10
- Calculations are explained _____/10

Recorded Presentation

_____/30

- The video presentation does not exceed 5 min. time limit; the presentation is professionally delivered, with clear audio, eye contact. The face of the presenter(s) appear(s) often. _____/10
- The presentation is clear and concise and answers all aspect (problem statement, solution, results discussion) _____/20

TOTAL

_____/100